

Homework 5

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

- 8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

Question 8.6

- a. Test the null hypothesis $H_0: \sigma_M^2 = \sigma_F^2$ against $H_A: \sigma_M^2 \neq \sigma_F^2$ at the 5% level.

```
SSE_M <- 97161.9174
N_M <- 577
K <- 4 # WAGE = b1 + b2 EDUC + b3 EXPER + b4 METRO + e
df_M <- N_M - K
var_M <- SSE_M / df_M

sigma_F_hat <- 12.024
var_F <- sigma_F_hat^2
N_F <- 1000 - 577
df_F <- N_F - K

F_stat_a <- var_M / var_F
F_crit_a_lower <- qf(0.025, df_M, df_F)
```

```

F_crit_a_upper <- qf(0.975, df_M, df_F)
p_val_a <- 2 * min(pf(F_stat_a, df_M, df_F), 1 - pf(F_stat_a, df_M, df_F))

cat("var_M =", var_M, "\nvar_F =", var_F, "\nF-stat =", F_stat_a, "\n")

## var_M = 169.567
## var_F = 144.5766
## F-stat = 1.172853

```

- **Test Statistic:** $F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = 1.173$
- **Rejection Region:** $F < 0.838$ or $F > 1.197$
- **Conclusion:** Since $F = 1.173$ is within the non-rejection region (p-value $\approx 0.082 > 0.05$), we do not reject the null hypothesis. There is no significant evidence that the variance of wages differs between males and females.

b. Specify the null hypothesis $H_0: \sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus $H_A: \sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$ at the 5% level.

```

SSE_S <- 56231.0382
N_S <- 400
K_b <- 5 # WAGE = b1 + b2 EDUC + b3 EXPER + b4 METRO + b5 FEMALE + e
df_S <- N_S - K_b
var_S <- SSE_S / df_S

SSE_M_b <- 100703.0471
N_M_b <- 600
df_M_b <- N_M_b - K_b
var_M_b <- SSE_M_b / df_M_b

F_stat_b <- var_M_b / var_S
F_crit_b <- qf(0.95, df_M_b, df_S)
p_val_b <- 1 - pf(F_stat_b, df_M_b, df_S)

cat("var_S =", var_S, "\nvar_M_b =", var_M_b, "\nF-stat =", F_stat_b, "\n")

## var_S = 142.3571
## var_M_b = 169.2488
## F-stat = 1.188904

```

- **Test Statistic:** $F = \frac{\hat{\sigma}_{MARRIED}^2}{\hat{\sigma}_{SINGLE}^2} = 1.189$
- **Rejection Region:** $F > F_{(0.95, 595, 395)} = 1.165$
- **Conclusion:** Since $F = 1.189 > 1.165$, we reject H_0 . There is sufficient evidence to support the claim that wages for married individuals are more variable than for single individuals.

c. NR^2 test conclusion.

- **Test Statistic:** $LM = 59.03$
- **Degrees of freedom:** The auxiliary regression contains $S = 4$ variables (EDUC, EXPER, METRO, FEMALE). Critical value $\chi^2_{(0.95,4)} = 9.488$.
- **Conclusion about heteroskedasticity:** Since $59.03 > 9.488$, we strongly reject the null hypothesis of homoskedasticity.
- **Evidence about part (b):** No. The right-hand side variables of (XR8.6b) do not include the MARRIED variable. Therefore, this NR^2 test assesses heteroskedasticity related to education, experience, gender, and living area, not marital status.

d. White test.

- **Variables:** The test includes the original 4 variables (excluding intercept), their squares (only continuous variables EDUC and EXPER have meaningful squares), and all their cross-products. Number of terms $S = 4 + 2 + 6 = 12$.
- **Degrees of freedom:** $df = 12$
- **5% Critical Value:** $\chi^2_{(0.95,12)} = 21.026$
- **Conclusion:** Since $LM = 78.82 > 21.026$, we reject the null hypothesis. There is strong evidence of heteroskedasticity.

e. Comparing OLS vs Robust Standard Errors.

- **Narrower interval estimates (smaller std. error):** EXPER (0.031 $\rightarrow$ 0.029), METRO (1.05 $\rightarrow$ 0.81), FEMALE (0.84 $\rightarrow$ 0.80).
- **Wider interval estimates (larger std. error):** EDUC (0.14 $\rightarrow$ 0.16).
- **Inconsistency?** No. Robust standard errors can be either larger or smaller than OLS standard errors. It depends on the correlation between the squared residuals and the squared explanatory variables.

f. Married t-value = 1.0. Does this conflict with (b)?

- **Conflict?** No.
- **Explanation:** Part (b) tests for heteroskedasticity, which is whether the *variance* of the error term depends on marital status. The t-test in part (f) tests whether the *conditional mean* of the wage depends on marital status. These evaluate two different parts of the model distribution, so there is no conflict.

Question 8.16

a. 95% CI for the effect of one more child on miles traveled.

```
data("vacation")
mod_a <- lm(miles ~ income + age + kids, data = vacation)
summary(mod_a)$coefficients
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	-391.54801	169.775215	-2.306273	2.214108e-02

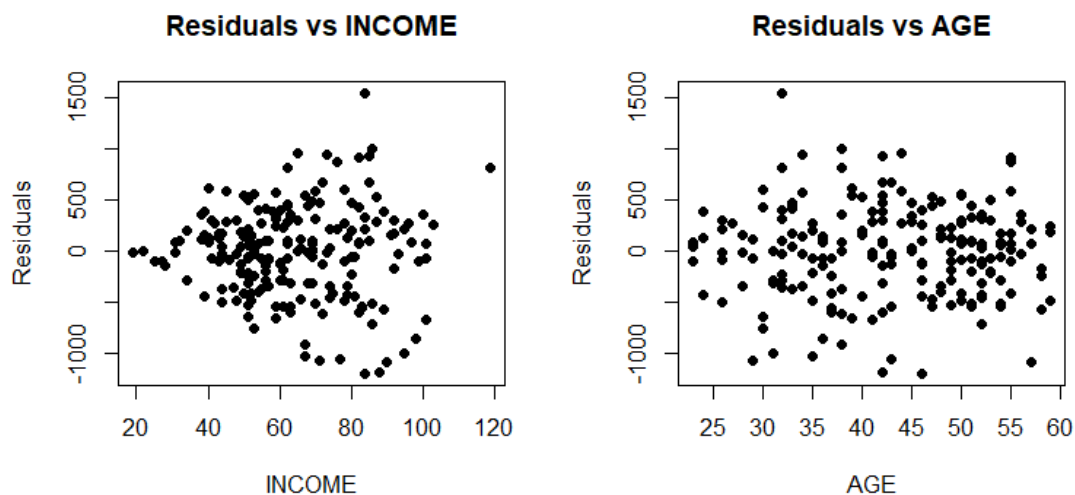
```
## income      14.20133    1.800256    7.888506 2.100345e-13
## age         15.74092    3.757370    4.189346 4.227760e-05
## kids       -81.82642   27.129596   -3.016131 2.899464e-03

b4 <- coef(mod_a)["kids"]
se4 <- sqrt(diag(vcov(mod_a))["kids"])
t_crit <- qt(0.975, mod_a$df.residual)

ci_lower_a <- b4 - t_crit * se4
ci_upper_a <- b4 + t_crit * se4
cat("95% CI (OLS): [", ci_lower_a, ", ", ci_upper_a, "]\n")

## 95% CI (OLS): [ -135.3298 , -28.32302 ]
```

b. Plot OLS residuals versus INCOME and AGE.



Observation: The plot of residuals against INCOME shows a distinct “fan” or “cone” shape (dispersion increases as income increases). This suggests the presence of heteroskedasticity. The AGE plot does not show a strong pattern.

c. Goldfeld-Quandt test.

```
vacation_sorted <- vacation[order(vacation$income),]
mod_c_lower <- lm(miles ~ income + age + kids, data = vacation_sorted[1:
90, ])
mod_c_upper <- lm(miles ~ income + age + kids, data = vacation_sorted[1
11:200, ])

var_lower <- sum(residuals(mod_c_lower)^2) / mod_c_lower$df.residual
var_upper <- sum(residuals(mod_c_upper)^2) / mod_c_upper$df.residual

F_stat_c <- var_upper / var_lower
df_lower <- mod_c_lower$df.residual
```

```
df_upper <- mod_c_upper$df.residual
F_crit_c <- qf(0.95, df_upper, df_lower)
```

- **Null Hypothesis:** $H_0: \sigma_{upper}^2 = \sigma_{lower}^2$ (homoskedasticity)
- **Alternative Hypothesis:** $H_A: \sigma_{upper}^2 > \sigma_{lower}^2$ (heteroskedasticity related to income)
- **Test Statistic:** $F = \frac{315821.6}{101744.6} = 3.104$
- **Rejection Region:** $F > F_{(0.95, 86, 86)} = 1.429$
- **Conclusion:** We reject H_0 . There is strong evidence that error variance is higher for larger incomes.

d. Robust 95% CI.

```
cov_hc1 <- vcovHC(mod_a, type = "HC1")
se_hc1 <- sqrt(diag(cov_hc1))["kids"]
ci_lower_d <- b4 - t_crit * se_hc1
ci_upper_d <- b4 + t_crit * se_hc1
cat("95% CI (Robust): [", ci_lower_d, ", ", ci_upper_d, "]\n")
## 95% CI (Robust): [ -139.323 , -24.32986 ]
```

- **Comparison:** The robust standard error is slightly smaller than the OLS standard error, resulting in a slightly narrower confidence interval ([-139.32, -24.33]) compared to part (a) ([-135.33, -28.32]).

e. GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$.

```
mod_e <- lm(miles ~ income + age + kids, data = vacation, weights = 1/income^2)
b4_e <- coef(mod_e)["kids"]
se4_e <- sqrt(diag(vcov(mod_e))["kids"])
t_crit_e <- qt(0.975, mod_e$df.residual)

ci_lower_e_conv <- b4_e - t_crit_e * se4_e
ci_upper_e_conv <- b4_e + t_crit_e * se4_e

cov_hc1_e <- vcovHC(mod_e, type = "HC1")
se_hc1_e <- sqrt(diag(cov_hc1_e))["kids"]
ci_lower_e_rob <- b4_e - t_crit_e * se_hc1_e
ci_upper_e_rob <- b4_e + t_crit_e * se_hc1_e

cat("GLS Conv 95% CI: [", ci_lower_e_conv, ", ", ci_upper_e_conv, "]\n")
## GLS Conv 95% CI: [ -119.8945 , -33.71808 ]

cat("GLS Robust 95% CI: [", ci_lower_e_rob, ", ", ci_upper_e_rob, "]\n")
## GLS Robust 95% CI: [ -121.4134 , -32.19919 ]
```

- **Comparison:** The conventional GLS interval $[-119.89, -33.72]$ and the robust GLS interval $[-121.41, -32.20]$ are both narrower than the OLS interval. Both intervals are quite similar, indicating that the assumed variance structure ($\sigma_i^2 = \sigma^2 INCOME_i^2$) captures most of the heteroskedasticity.